

CASD: Chunk-Aligned Semantic Distillation for Multi-Stage Robot Manipulation

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Abstract

An action chunk can span several stages of a manipulation task, yet a label for its first step describes only the current stage. We introduce Chunk-Aligned Semantic Distillation (CASD), which derives semantic targets for entire action chunks. An offline vision–language model segments demonstrations into described stages. Their occupancy within each action chunk determines a weighted semantic target, including transitions between stages. A CASD generator learns to predict this target from the current observation, robot state, and task instruction. We then freeze the generator and train a policy conditioned on its predictions. The semantic branch runs once per policy query, without online VLM calls or reasoning-trace decoding. Teacher matching on annotated LIBERO training episodes is above chance for both single-stage and boundary-crossing chunks. We evaluate three Fast-WAM variants and a DreamZero integration across four benchmarks, including distribution shifts on LIBERO-Plus. Compared with published references, IDM+CASD reaches 98.9% versus 98.0% average success on LIBERO, while Uncond falls below its reference. Joint+CASD reaches 93.0% versus 90.6% on RoboTwin 2.0, and DreamZero+CASD reaches a 47.9% four-category MolmoSpaces manipulation average versus 40.7%. Performance varies across backbone integrations.

1 Introduction

Many vision–language–action (VLA) policies predict a chunk of future actions at each query (Zhao et al. 2025; Huang et al. 2026). World Action Models (WAMs) also model future visual states (Yuan et al. 2026; Ye et al. 2026). Chunking reduces the number of policy queries. In multi-stage tasks, however, the immediate objective can change within a predicted chunk while the overall instruction stays fixed. A label for the current step may then omit stages that the chunk will reach.

Consider placing a bowl in a drawer and then closing it. A chunk near the placement boundary may contain transporting the bowl, releasing it, and reaching toward the drawer handle. Labeling that chunk only as *transport* omits the upcoming release; labeling it only as *close* skips an unfinished interaction. Even a list of the overlapping stages leaves out how much of the chunk each stage occupies. This motivates

a semantic target defined over the same time interval as the actions it supervises.

Language-conditioned action hierarchies (Belkhale et al. 2024) and progress models (Chen et al. 2025a; Yan et al. 2026; Feng et al. 2026) provide temporal structure for control. Visual chain-of-thought (CoT) and latent planning use intermediate goals or plans to guide actions (Zhao et al. 2025; Huang et al. 2026), while privileged distillation transfers demonstrated future information to a current-only policy (Fang, Chen, and Cai 2026). We focus on supervision at the policy’s prediction timescale, describing both the stages that overlap an action chunk and the fraction of the chunk occupied by each.

A segmented demonstration supplies the required supervision. For a query at time t , the fraction of its valid future action chunk spent in each stage defines a chunk-level context: a single stage away from a boundary and a weighted mixture near one. The weights depend on the demonstrated future, so they are privileged information unavailable during deployment.

We transfer this context to the policy through Chunk-Aligned Semantic Distillation (CASD). An offline VLM-plus-action pipeline divides training trajectories into contiguous stages with free-form descriptions. We encode the descriptions and mix their semantic representations according to their chunk occupancies. A CASD generator predicts this target from the current observation, robot state, and task instruction. The target specifies within-horizon stage composition, using free-form descriptions without a global vocabulary of stage classes.

Stage A trains the CASD generator against this fixed teacher, separately from the policy. Stage B freezes it and trains the video–action policy on its predictions. Cross-attention adapters supply these tokens to the backbone. The policy consequently receives predicted semantic conditions during both training and deployment. The generator predicts all semantic slots in parallel once per policy query; its branch makes no online VLM calls and generates no textual reasoning trace or subgoal image.

We evaluate whether the generator recovers the teacher context and whether that context helps execution. Teacher matching on annotated LIBERO training episodes is above chance for both single-stage and boundary-crossing chunks. Policy evaluation covers three Fast-WAM variants, distribu-

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tion shifts on LIBERO-Plus, bimanual tasks on RoboTwin, and MolmoSpaces with DreamZero. The LIBERO average improves for IDM and Joint but declines for Uncond.

Our contributions are:

- Chunk-level semantic supervision that weights stage descriptions by their occupancy in the action horizon, including chunks that cross stage boundaries.
- A current-input predictor and policy-conditioning interface that transfer this supervision to control, with the same frozen predictor supplying semantic conditions during policy learning and deployment.
- Evaluations of teacher matching, three Fast-WAM variants, distribution shifts on LIBERO-Plus, bimanual tasks on RoboTwin, and a DreamZero integration, including settings where performance declines.

2 Related Work

Stages, subgoals, and progress. Hierarchical control uses grounded skills (Ahn et al. 2022), executable programs (Liang et al. 2022), and language/video plans (Ajay et al. 2023). RT-H predicts fine-grained language motions from observations and the task instruction, then conditions action prediction on these motions within the same imitation policy (Belkhale et al. 2024). PALO searches over language decompositions and ordered trajectory partitions, using a pretrained policy’s action-reconstruction error to choose conditions that fit the target demonstrations (Myers et al. 2025). PERIA combines language sub-instructions with imagined visual subgoals (Ni et al. 2024). UVD discovers visual subgoals by detecting changes in pretrained visual embeddings along demonstrations (Zhang et al. 2023), and Long-VLA selects visual context using motion and interaction phases (Fan et al. 2025). SARM, ProgressVLA, and ProcVLM study stage-aware rewards or progress supervision (Chen et al. 2025a; Yan et al. 2026; Feng et al. 2026). CASD weights free-form stage descriptions by their occupancy in the action chunk. The target describes the composition of the prediction horizon: queries with the same current-stage label can receive different targets, and changing the horizon can change the target at the same observation. This places semantic supervision at the timescale of action prediction, without requiring a global stage vocabulary. The distinction is the supervision target: stage composition within the action horizon, rather than a scalar measure of episode completion.

Plans and reasoning for control. CoT-VLA autoregressively predicts future images as visual subgoals (Zhao et al. 2025), while ThinkAct learns visual latent plans (Huang et al. 2025). Fast-ThinkAct distills linguistic reasoning and visual trajectory information into compact latents, with spatial tokens that predict trajectory waypoints in parallel (Huang et al. 2026). ACoT-VLA uses action-space reference trajectories to guide action prediction (Zhong et al. 2026). ERVLA uses reasoning dropout to learn from embodied CoT without decoding it during action prediction (Sun et al. 2026); ZR-0 likewise trains an action expert that can bypass reasoning-trace generation (Li et al. 2026a). CASD directly predicts semantic context once per query, without online VLM calls,

text decoding, or subgoal-image generation in this branch. Its teacher specifies which interactions overlap the action chunk and their relative durations. It omits explicit paths and stage order; the policy also receives current observations, robot state, and the task instruction. Identical descriptions and occupancies yield the same teacher even when execution order differs.

Privileged distillation. PFD transfers future information through the difference between future-conditioned and current-only action-denoising outputs, fitting this residual with a student adapter (Fang, Chen, and Cai 2026). CASD constructs a fixed teacher from demonstrated stage descriptions and occupancies, independently of policy-teacher action outputs. The generator predicts this target from current inputs and is then frozen for policy training. Fast-ThinkAct also freezes its learned planner (Huang et al. 2026); freezing and parallel token prediction are shared design choices. The supervision differs: CASD distills stage-duration mixtures rather than reasoning traces or trajectories. Training the policy on generator predictions exposes it to the same conditioning source, including prediction errors, that it receives at deployment. Demonstrated future boundaries are used only to construct training targets; they are not inputs to the deployed generator.

World models for action prediction. OpenVLA learns observation-to-action mappings through robot pretraining (Kim et al. 2025). WAMs additionally learn visual dynamics: Motus and LingBot-VA couple video and action modeling (Bi et al. 2025; Li et al. 2026b), and DreamZero adapts a pretrained video-diffusion backbone for control (Ye et al. 2026). Fast-WAM separates video co-training from future-video generation at inference (Yuan et al. 2026). Its Joint and IDM variants denoise video and actions together or sequentially, respectively, while Uncond uses current-frame features. RIFT constructs future-conditioning features in one pass (Zhang et al. 2026). CASD adds semantic conditioning while retaining the backbone objectives and generation procedure. In Fast-WAM, both experts receive the predicted tokens through separate projections and adapters. The generator runs once per query, and adapters run at each denoising step. The action policy retains responsibility for selecting the executed sequence.

3 Method

3.1 Chunk-Aligned Supervision

We consider demonstrations pairing a task instruction ℓ with a trajectory $\{(o_t, s_t, a_t)\}_{t=0}^{T-1}$ of observations, robot states, and actions. The policy predicts an action chunk from $x_t = (o_t, s_t, \ell)$. We first define its semantic supervision independently of the backbone, then describe prediction and policy conditioning. Figure 1 summarizes the two training stages.

During training, we use GPT-5.4 as an offline VLM annotator (OpenAI 2026). Given the task instruction, ordered frames, extracted task entities, and action-derived boundary hints, it divides the episode into an ordered, non-overlapping

sequence of action-centered stages. Deterministic postprocessing enforces full coverage and refines gripper transitions. We write the semantic annotation as $\{([b_k, e_k], u_k)\}_{k=1}^K$, where u_k is a free-form stage description. Each policy query at time t is paired with an H -step demonstration action chunk. Let $\mathcal{C}_t$ contain its valid, non-padded steps, and let g_τ be the annotated stage at step τ . We define the chunk occupancy of stage k as

$$w_{t,k} = \frac{|\{\tau \in \mathcal{C}_t : g_\tau = k\}|}{|\mathcal{C}_t|}, \quad \sum_k w_{t,k} = 1. \quad (1)$$

For Fast-WAM, we use $H = 32$ and include every annotated stage that overlaps the valid chunk. The target represents all stages in the chunk. For example, a 32-step chunk with 24 transport actions and eight release actions has occupancies $3/4$ and $1/4$. A chunk fully within transport has occupancy one for that stage. Near the episode end, the denominator counts only the remaining valid actions. These occupancies use the demonstrated future and serve as privileged training labels.

3.2 Chunk-Aligned Semantic Teacher

Let the frozen UMT5 encoder (Chung et al. 2023) produce the valid token sequence $E(u_k) = \{h_{k,j}\}_{j=1}^{L_k}$ for stage description u_k . We represent each description with N ordered slots by partitioning the sequence into contiguous, non-empty intervals $\{\mathcal{I}_{k,n}\}_{n=1}^N$ and mean-pool each interval:

$$\bar{h}_{k,n} = \frac{1}{|\mathcal{I}_{k,n}|} \sum_{j \in \mathcal{I}_{k,n}} h_{k,j}. \quad (2)$$

The slots follow relative positions in the text sequence, independently of the number of stages in the chunk. Let $\mu \in \mathbb{R}^{D_e}$ be the mean of pooled training-description vectors and $R \in \mathbb{R}^{D_p \times D_e}$ a fixed dimension-reduction map. The occupancy-weighted teacher for slot n is

$$T_{t,n} = R \left(\sum_k w_{t,k} \bar{h}_{k,n} - \mu \right), \quad T_t \in \mathbb{R}^{N \times D_p}. \quad (3)$$

Because the occupancies sum to one, this equals the weighted mixture of the per-stage projected tokens. Each target has the same size regardless of how many stages overlap the chunk. The mixture records stage proportions but does not encode their order. The policy also receives the current observation and instruction. The teacher is fixed before student learning. Token count and projection settings are specified in the experimental setup, with full construction details in the supplement.

3.3 CASD Generator from Current Inputs

The CASD generator maps the current visual latent z_t , robot state s_t , and task instruction to P_t . A frozen VAE encodes the current frame, and an observation MLP fuses its features with the robot state. This feature and the projected mean-pooled instruction form a conditioning vector c_t . The model produces N parallel queries from c_t and attends to the full

instruction sequence $H^\ell = E(\ell)$:

$$\begin{aligned} Q_t &= \text{reshape}(W_q c_t), \\ K^\ell &= V^\ell = W_{kv} H^\ell, \\ A_t &= \text{MHA}(Q_t, K^\ell, V^\ell; m), \\ P_t &= F_{\text{tok}}(Q_t + A_t). \end{aligned} \quad (4)$$

Here m masks padded instruction tokens, and F_{tok} is a layer normalization followed by a linear projection. Adding Q_t to A_t lets visual and proprioceptive features influence the output directly, rather than only through text-attention weights. The N queries are processed in parallel, and each corresponds to an ordered semantic slot in Equation (2). The generator predicts the mixed target directly.

3.4 Stage-A Semantic Distillation

Stage A trains only the CASD generator, sampling frame anchors from annotated training demonstrations. The VAE, UMT5 encoder, teacher transformation, and policy remain frozen. Precomputed current-frame features use the same VAE and preprocessing as the online path.

Let σ_T denote the RMS scale stored with the teacher. Because teacher and prediction slots have a fixed correspondence, we use normalized direct regression:

$$d(P, T) = \frac{\|P - T\|_F^2}{ND_p \sigma_T^2}, \quad \mathcal{L}_{\text{dir}} = \frac{\sum_i \beta_i d(P_i, T_i)}{\sum_i \beta_i}. \quad (5)$$

Here $\beta_i \in \{0, 1\}$ indicates whether sample i has a non-padded action step and passes the episode-length and stage-boundary checks. All retained samples have unit weight; stored annotation confidence and validation flags do not affect this mask.

To separate contexts within the same task, we consider other valid teacher candidates T_j from that task in the global minibatch, with $d(T_i, T_j) \geq \delta$ and choose the one that maximizes the ranking violation $d(P_i, T_i) - d(P_i, T_j) + \rho d(T_i, T_j)$. Let T_i^- denote this candidate and $\mathcal{I}^-$ the samples for which one exists. With $d_i^+ = d(P_i, T_i)$, $d_i^- = d(P_i, T_i^-)$, and $\Delta_i = d(T_i, T_i^-)$, the ranking objective is

$$\mathcal{L}_{\text{rank}} = \frac{\sum_{i \in \mathcal{I}^-} \beta_i [d_i^+ - d_i^- + \rho \Delta_i]_+}{\sum_{i \in \mathcal{I}^-} \beta_i}. \quad (6)$$

The Stage-A objective is $\mathcal{L}_A = \mathcal{L}_{\text{dir}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}$. We use $\lambda_{\text{rank}} = 1$, $\delta = 0.05$, and $\rho = 0.5$. Each loss is zero when its denominator is zero.

3.5 Stage-B Policy Learning and CASD Injection

Stage B loads the Stage-A checkpoint, freezes the CASD generator, and conditions the policy on its predictions. The same token source is used during policy training and deployment. Teacher targets and alignment losses are not used in Stage B; the objective remains the original sum of visual-dynamics and action flow-matching losses,

$$\mathcal{L}_B = \mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{act}}. \quad (7)$$

In the Fast-WAM implementation (Yuan et al. 2026), the video and action experts share predicted tokens P_t and

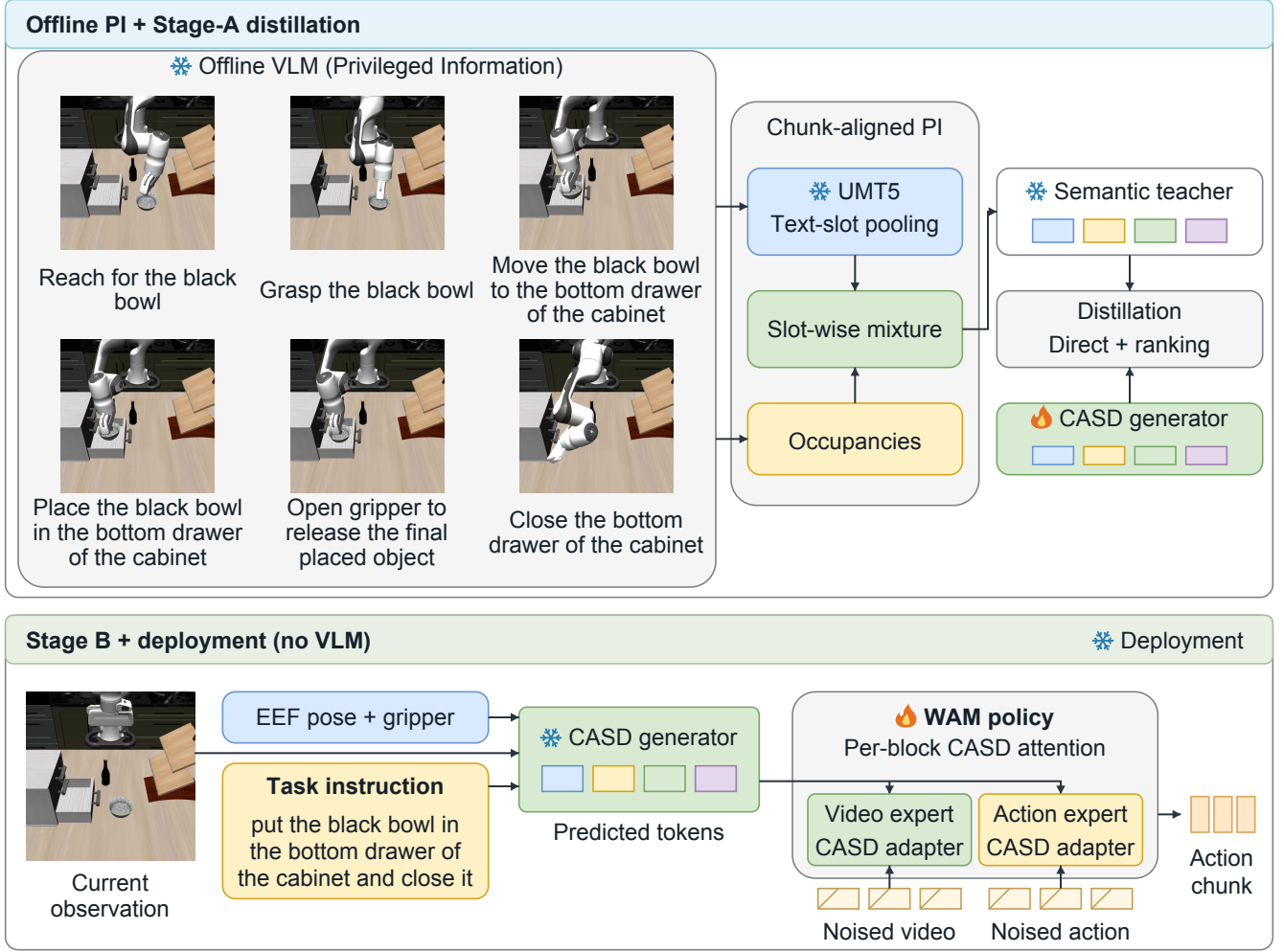


Figure 1: Two-stage privileged semantic distillation in CASD. Offline stage annotations and chunk occupancies define a fixed semantic teacher. Stage A trains the CASD generator to predict these tokens from current inputs. Stage B freezes the generator and conditions the video-action policy on its predictions. The offline annotations and future-dependent targets are absent at deployment.

project them separately to $C_{t,e} = g_e(P_t) \in \mathbb{R}^{N \times D_c}$. Each g_e is an independently learned affine map followed by GELU. Each block receives a separately parameterized low-rank cross-attention adapter:

$$\tilde{X}_e^{(l)} = X_{e,\text{std}}^{(l)} + \mathcal{A}_{p,e}^{(l)}(X_e^{(l)}, C_{t,e}), \quad e \in \{v, a\}. \quad (8)$$

Policy parameters and adapters are optimized under the backbone losses while the CASD generator stays frozen. We use $D_c = 1024$ and rank-192 adapter bottlenecks. The final adapter projection is initialized to zero, preserving the pretrained function at initialization. Full architecture dimensions are provided in the supplementary material.

3.6 Inference

Before each diffusion loop, the CASD generator computes the context once from the latest visual latent, current robot state, and task instruction. The policy reuses this context

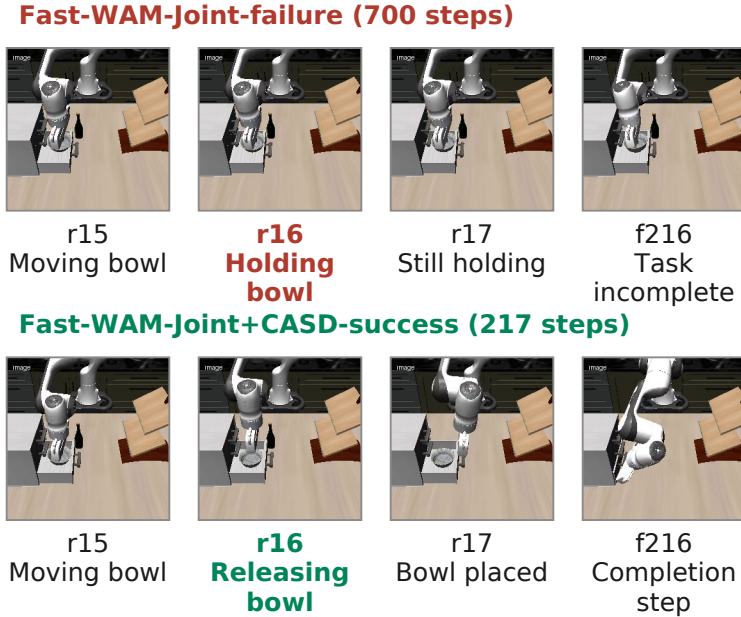
across diffusion steps and recomputes it after the environment advances. The original backbone determines how video and action are denoised: Joint generates them together, IDM generates video before actions, and Uncond uses current-frame features. Across variants, CASD predicts all semantic slots in parallel once per policy query, without online VLM calls, text decoding, or subgoal-image generation. CASD adapters run at each denoising step.

4 Experiments

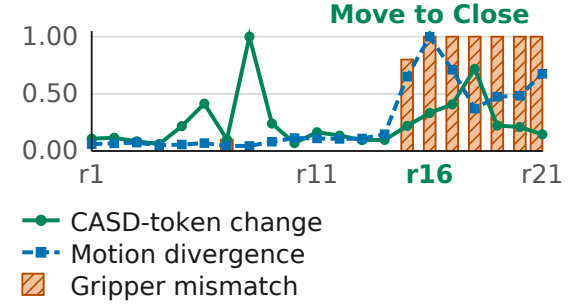
4.1 Experimental Setup

Benchmarks. We evaluate on four benchmarks. LIBERO contains Spatial, Object, Goal, and Long suites (Liu et al. 2023); LIBERO-Plus applies seven controlled shifts spanning layout, camera, robot initialization, language, lighting, background, and sensor noise (Fei et al. 2026). RoboTwin 2.0 evaluates 50 bimanual tasks under clean and domain-

(a) Paired Rollout Divergence



(b) Signal Timeline



(c) Semantic Transition at r16

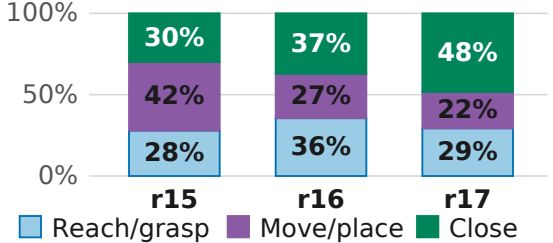


Figure 2: A paired LIBERO-Long rollout showing where the baseline and CASD trajectories diverge. The Fast-WAM-Joint baseline times out, whereas Fast-WAM-Joint+CASD succeeds from the same recorded initial observation with matched inference seeds. (a) By replan r16, CASD has begun releasing the bowl while the baseline keeps holding it. (b) This behavioral difference coincides with the largest motion divergence and complete gripper-command disagreement; token and motion signals are normalized separately. (c) A post-hoc convex fit to task-specific teacher prototypes shows a shift from Move/place toward Close. The displayed weights describe the fit and do not enter the policy.

randomized conditions (Chen et al. 2025b). MolmoSpaces-Bench spans manipulation and navigation in diverse indoor scenes (Kim et al. 2026). We evaluate the four-category Franka/DROID manipulation subset used by the leaderboard snapshot: Pick, Pick-and-Place, Open, and Close (Allen Institute for AI 2026). Policy tables report success rates (%); Table 1 reports teacher-matching recall on a $[0, 1]$ scale.

Model setup. On LIBERO and LIBERO-Plus, we evaluate CASD with the Uncond, Joint, and IDM Fast-WAM variants. RoboTwin 2.0 includes the same three CASD variants, alongside the three official Fast-WAM baselines. These models build on Wan2.2-TI2V-5B (Yuan et al. 2026; Wan Team 2025). The CASD runs use a frozen 10,000-step Stage-A CASD generator and a fixed seed-0 random-projection teacher with $N = 4$ tokens of dimension $D_p = 128$, with CASD routed to both video and action experts.

For cross-backbone evaluation, we integrate CASD into DreamZero (Wan2.1-I2V-14B) (Ye et al. 2026); both its Franka pretraining and our CASD-augmented training use DROID (Khazatsky et al. 2024). Our CASD policies use predicted semantic conditions in Stage B, matching the context source at evaluation. Tables 2 and 4 compare our runs with published results for the corresponding backbone variants and benchmark settings. The LIBERO-Plus references are separately evaluated checkpoints reported by RIFT. These

comparisons do not control for differences in training runs. We express differences in percentage points (pp). The supplement provides the LIBERO pipeline, metric definitions, and result sources.

Subset	Queries	Share	Chance Recall	CASD Recall
Pure	5,312	55.87%	0.185	0.430
Mixed	4,195	44.13%	0.174	0.765
Mixed-2	3,643	38.32%	0.175	0.761
Mixed-3+	552	5.81%	0.166	0.790

Table 1: Episode-local teacher matching at annotated training phase starts. Pure queries cover one stage in the valid part of the following 32-step chunk; Mixed queries cross stage boundaries. Both recalls are query averages; Mixed-2 and Mixed-3+ partition Mixed.

4.2 Teacher Matching from Current Inputs

Using only current inputs, we evaluate the frozen CASD generator on annotated LIBERO training episodes. After prediction, each query is matched against teacher targets at annotated phase starts in the same episode, ranked by negative normalized MSE. Recall@1 assigns credit $1/m$ when the correct target is among m candidates tied for the best

Method	Camera	Robot Init.	Language	Light	Background	Noise	Layout	Pooled
NORA	2.2	37.0	65.1	45.7	58.6	12.8	62.1	39.0
WorldVLA	0.1	27.9	41.6	43.7	17.1	10.9	38.0	25.0
UniVLA	1.8	46.2	69.6	69.0	81.0	21.2	31.9	42.9
π_0	13.8	6.0	58.8	85.0	81.4	79.0	68.9	53.6
π_0 -FAST	65.1	21.6	61.0	73.2	73.2	74.4	68.8	61.6
RIPT-VLA	55.2	31.2	77.6	88.4	91.6	73.5	74.2	68.4
OpenVLA-OFT-M	55.6	21.7	81.0	92.7	91.0	78.6	68.7	67.9
Fast-WAM [†]	16.5	43.4	67.8	80.1	51.6	37.9	61.2	49.7
Fast-WAM+CASD	28.8	49.9	75.2	83.8	54.5	51.0	64.8	57.2
Fast-WAM-Joint [†]	38.7	63.9	93.1	94.7	55.7	57.7	77.5	68.1
Fast-WAM-Joint+CASD	46.0	70.7	94.5	94.5	62.3	60.3	81.3	72.2
Fast-WAM-IDM [†]	46.2	69.1	94.2	92.8	56.4	61.8	81.5	71.4
Fast-WAM-IDM+CASD	50.3	77.0	90.6	92.8	67.1	62.8	81.2	73.9

Table 3: LIBERO-Plus success scores (%). Pooled is the variant-micro rate over 10,030 benchmark variants; external comparison rows follow the benchmark report (Fei et al. 2026). [†] RIFT-reported Fast-WAM references (Zhang et al. 2026); CASD rows are ours.

score, and zero otherwise. Chance Recall averages $1/N_e$ over queries, where N_e is the episode gallery size.

The evaluation contains 9,507 phase-start queries from 1,712 episodes (5.55 per episode). Queries are grouped by the stages in their valid following 32-step chunk, as defined in Table 1. Recall is $2.3\times$ chance for Pure queries (0.430 vs. 0.185) and $4.8\times$ for Mixed-3+ (0.790 vs. 0.166). These training-set diagnostics show recovery of boundary-crossing contexts from current inputs; comparisons between subsets also depend on the distinctiveness of their gallery targets.

4.3 Downstream Policy Evaluation

In-distribution performance on LIBERO. Table 2 shows average differences of -3.2 , $+0.3$, and $+0.9$ pp for Uncond, Joint, and IDM, respectively, using unrounded run averages. IDM+CASD reaches the highest average in the table (98.9%). Joint improves from 98.5% to 98.8%, with its largest gain on Long (+1.2 pp), followed by Goal (+0.4 pp); Spatial and Object change by -0.2 and -0.4 pp. The Uncond decline is largest on Goal (-4.4 pp). Two of the three CASD variants score above their published references on average, with gains and regressions across individual suites.

Illustrative paired case. Figure 2 examines trial 2 of “put the black bowl in the bottom drawer of the cabinet and close it.” The Fast-WAM-Joint baseline succeeds in 49/50 trials and CASD in 50/50; trial 2 is their paired failure/success case, with the same recorded initial observation and matched seeds at all 22 common replans.

By replan r16, CASD has begun releasing the bowl while the baseline keeps holding it. The r15→r16 transition has token change 0.060, non-gripper action RMSE 0.460, and disagreement on all 10 aligned gripper commands. In the visualization-only fit, Move/place falls from 42% to 27% and then 22%, while Close rises from 30% to 37% and 48%. CASD begins releasing at policy step 152 and completes the task at zero-indexed step 216 (217 executed steps); the baseline releases 84 steps later and times out at 700 steps.

This semantic transition coincides with the largest motion divergence among the 22 common replans.

Method	Emb. PT	Spatial	Object	Goal	Long	Avg.
$\pi_{0.5}$	✓	98.8	98.2	98.0	92.4	96.9
GR00T-N1.7	✓	97.7	98.5	97.5	94.4	97.0
Motus	✓	96.8	99.8	96.6	97.6	97.7
Qwen-RM-scratch	×	—	—	—	—	98.2
LingBot-VA	✓	98.5	99.6	97.2	98.5	98.5
Fast-WAM [†]	×	98.2	100.0	97.0	95.2	97.6
Fast-WAM+CASD	×	97.6	94.8	92.6	92.8	94.5
Fast-WAM-Joint [†]	×	99.6	99.4	98.2	96.8	98.5
Fast-WAM-Joint+CASD	×	99.4	99.0	98.6	98.0	98.8
Fast-WAM-IDM [†]	×	98.8	97.8	97.8	97.6	98.0
Fast-WAM-IDM+CASD	×	99.6	99.8	98.2	97.8	98.9

Table 2: LIBERO success rates (%). Emb. PT: embodied pretraining; Qwen-RM: Qwen-RobotManip; unsuffixed Fast-WAM: Uncond. Bold marks the column best. [†] marks official Fast-WAM results (Yuan et al. 2026). External rows follow $\pi_{0.5}$ (Physical Intelligence et al. 2025), GR00T-N1.7 (NVIDIA 2026), Motus (Bi et al. 2025), Qwen-RM (Qwen Team 2026), and LingBot-VA (Li et al. 2026b).

Robustness under LIBERO-Plus shifts. Table 3 reports CASD-augmented Fast-WAM results under seven controlled visual, robot-state, and language shifts.

Our CASD runs reach 57.2%, 72.2%, and 73.9% pooled success for Fast-WAM, Joint, and IDM, respectively. RIFT reports external references of 49.7%, 68.1%, and 71.4%. Because the checkpoints and training recipes differ, the $+7.5/+4.1/+2.5$ pp differences are cross-report comparisons. Our Joint model is higher on six of seven shifts, with the largest gaps for camera viewpoint (+7.3 pp), robot initial state (+6.8 pp), and background texture (+6.6 pp). Camera viewpoint remains its weakest category (46.0%), well below language and lighting (both 94.5%). IDM+CASD has

the highest pooled rate among our variants, although its language score is lower than Joint+CASD.

Method	Embodied PT	Clean	Rand.	Average
$\pi_{0.5}$	✓	82.7	76.8	79.8
Motus	✓	88.7	87.0	87.8
Qwen-RM-scratch	×	88.7	88.4	88.6
LingBot-VA	✓	92.9	91.6	92.2
Fast-WAM [†]	×	91.9	91.8	91.8
Fast-WAM+CASD	×	90.5	91.4	90.9
Fast-WAM-Joint [†]	×	90.8	90.3	90.6
Fast-WAM-Joint+CASD	×	92.7	93.3	93.0
Fast-WAM-IDM [†]	×	91.2	91.3	91.3
Fast-WAM-IDM+CASD	×	92.6	92.2	92.4

Table 4: RoboTwin 2.0 success rates (%) across 50 tasks. Clean and Rand. denote Easy and Hard; Embodied PT denotes embodied pretraining; Qwen-RM abbreviates Qwen-RobotManip. Bold marks the column best. [†] marks official Fast-WAM baselines from Yuan et al. (2026); other external sources are cited in the text.

Bimanual robustness on RoboTwin 2.0. We report mean success across 50 tasks under clean (Easy) and domain-randomized (Hard) evaluation. External results follow their source papers (Physical Intelligence et al. 2025; Bi et al. 2025; Qwen Team 2026; Li et al. 2026b).

Joint+CASD exceeds the official Joint baseline by 1.9 pp on Clean and 3.0 pp on Rand. (+2.4 pp average). Uncond+CASD reaches 90.9% average (90.5% Clean and 91.4% Rand.). IDM+CASD scores 92.6% Clean and 92.2% Rand. (92.4% average). Joint+CASD leads Table 4 in Rand. (93.3%) and average (93.0%).

Method	Pick	P&P	Open	Close	Avg.
PRTS-Droid	53.8	27.7	12.0	76.3	42.5
MolmoAct2 DROID	48.3	23.3	11.7	71.3	38.7
$\pi_{0.5}$ DROID	36.4	13.6	22.7	65.1	34.5
π_0 DROID	16.2	12.5	11.0	53.1	23.2
LAP-VLA	24.9	6.6	11.4	45.9	22.2
π_0 -FAST DROID	22.0	10.9	11.1	38.6	20.7
Psi-R2	73.1	41.5	9.0	62.2	46.5
DreamZero	52.1	25.6	24.9	60.3	40.7
DreamZero+CASD	59.7	36.1	35.1	60.8	47.9

Table 5: Success rates (%) on four MolmoSpaces manipulation categories. P&P denotes Pick-and-Place; navigation is excluded. External values follow the official leaderboard snapshot (Allen Institute for AI 2026); the benchmark is described by Kim et al. (2026). The average is computed over the four listed categories. Bold marks the best value in each column.

Integration into DreamZero. We integrate CASD into DreamZero and evaluate four MolmoSpaces manipulation categories; navigation is outside this comparison.

DreamZero+CASD reaches 47.9% average success, compared with 40.7% for the leaderboard DreamZero reference (+7.2 pp). The largest category differences are on Pick-and-Place (+10.5 pp) and Open (+10.2 pp). DreamZero+CASD has the highest average in Table 5.

4.4 Discussion

Semantic context at the action horizon. Teacher matching shows that CASD can recover chunk-level context on annotated training episodes. The paired rollout in Figure 2 illustrates how a predicted semantic transition can accompany a change in manipulation behavior.

Interaction with the policy backbone. Joint and IDM exceed their published LIBERO and RoboTwin references, whereas Uncond declines. All three exceed the LIBERO-Plus references. Semantic context may complement each backbone’s learned representations differently. Matched comparisons of current-stage labels, equal-weight and occupancy-weighted mixtures, and expert routing could identify which conditions benefit each variant; the present reference comparisons do not isolate these factors.

Richer temporal predictions. The current teacher retains stage proportions but discards stage order. Constructing separate mixtures over successive parts of the action horizon would preserve coarse order while keeping a bounded token budget. A second direction is to predict several possible semantic futures using a short observation history. This could distinguish repeated visits to similar states and avoid averaging incompatible contexts when several continuations are plausible. Varying both the prediction horizon and the replanning interval would test how much temporal detail control needs.

Learning from execution. Automatic annotations can contain grounding and boundary errors, and policy rollouts can depart from the demonstrations used for distillation. Controlled boundary perturbations during distillation could test sensitivity to annotation timing. Separately, annotating policy rollouts would allow semantic prediction errors to be measured during execution. Execution feedback could then guide refinement of the teacher or adaptation of the generator. These extensions should be evaluated jointly for task success and query latency, including generator and adapter costs under matched hardware and sampling settings.

5 Conclusion

CASD aligns semantic supervision with the time span of an action chunk. A fixed teacher combines stage descriptions according to their occupancy, and a generator predicts the resulting context from current inputs. Freezing the generator gives policy learning and deployment the same source of semantic conditions. Evaluations with three Fast-WAM variants and DreamZero demonstrate its use across video-action generation schemes, with task success depending on the backbone and benchmark. The central contribution is a way to connect demonstrated task structure to the temporal resolution of action prediction.

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Supplementary Material

These appendices detail the LIBERO annotation and distillation pipeline, teacher-matching protocol, policy metrics, per-task RoboTwin results, and the paired rollout in Figure 2.

A Offline Phase Annotation

A.1 Annotation Unit and Raw Episode Inputs

Each annotation covers one complete LIBERO training demonstration and its benchmark instruction (Liu et al. 2023). The process reads the four local sources in Table S1. The VLM receives the task text, a frame contact sheet, and action-derived event hints. Episode identifiers, raw actions, and the future $H=32$ action occupancy remain local; occupancy is computed after annotation for Stage-A supervision.

The output for an episode is a temporally ordered sequence

$$\mathcal{S} = \{([b_k, e_k], u_k, q_k, v_k)\}_{k=1}^K,$$

where $[b_k, e_k]$ is a half-open frame interval, u_k is a free-form, action-first description, $q_k \in [0, 1]$ is a stored annotation score initialized from the returned confidence or a postprocessing rule, and v_k is an audit flag. Intervals are normalized to be contiguous, non-overlapping, positive in length, and to cover the episode. The scores and flags are retained for auditing. They do not set training weights, and the scores are not calibrated probabilities.

A.2 VLM-Visible Request

Contact sheet. The sampler begins with 12 uniform indices from frame 0 through frame $T - 1$, including both endpoints. Selected chronological action-hint frames may replace at most four interior cells. This retains global visual coverage while adding observations near likely transitions. Every cell is numbered left-to-right as $i/12$. The percentage printed inside a cell is its ordinal position in the strip; after a hint replacement it should therefore be read as an ordering cue, not an exact source timestamp. The accompanying text hint gives the exact source frame and its true normalized percentage.

Text prompt. The same user message contains the complete selected task instruction; objects, destinations, and extra action verbs extracted from that string; and the chronological hint list. A hint is rendered only as `t=., frame ., gripper opens/closes, velocity peak, or stillness`. Hint scores, raw action values, and detection thresholds are omitted. The prompt defines visually observable transition cues, asks the boundary to be the first frame at which a gripper change, object lift-off, arm-direction reversal, or destination entry becomes visible, and asks for the fewest ordered stages needed to complete the task. Each action verb in the task instruction must be represented, with separate Place and Release descriptions for every pick-and-place operation. Descriptions should contain 5–12 words, start with an action verb, and preserve task nouns. Suggested phase counts are 2–4 for a simple single-verb task, 4–6 for one pick-and-place target, and 5–8 for a multi-verb task; these are guidance, not fixed classes.

The request uses one multimodal user message containing the prompt and base64-encoded JPEG. The launcher requests the `gpt-5.4` model alias (OpenAI 2026), temperature 0.1, and at most 800 completion tokens. Sampling seed, top- p , and schema mode use the service defaults. The client retries failed requests up to four times and adapts unsupported parameter names.

The requested response is:

```
{"phase_segments":[{"subtask": string,
"start_ratio": float, "end_ratio":
float,
"confidence": float}]}
```

The VLM returns the ordered descriptions, coarse normalized boundaries, and per-segment confidence. The deterministic postprocessor derives final frame intervals, validity flags, and Release-event metadata.

A.3 Action-Hint Extraction

For the seven-dimensional LIBERO action, the motion signal is the Euclidean norm of the first six coordinates. A *velocity peak* exceeds the episode’s 70th percentile and is strictly larger than the values one and two frames on either side. A *stillness* candidate has a ten-entry local mean below the 20th percentile; nearby candidates with gaps of at most eight frames are clustered and represented by a cluster center.

The gripper is considered open above 0.5 and closed otherwise. A transition is emitted only after the new Boolean state persists for three consecutive frames; its event location is the first frame of that confirmed run, and the initial stable state is not an event. For the default 12-frame request, at most eight gripper transitions, five velocity peaks, and four stillness centers are considered. Velocity peaks within ten frames of an already selected event and stillness centers within twelve frames are skipped. Candidates are then prioritized using internal scores 1.0, 0.6, and 0.45, respectively; candidates within six frames of a retained event are deduplicated, and at most 16 hints remain. These scores affect local selection only; they are separate from the returned confidence used to initialize q_k .

A.4 Generic Response Postprocessing

The parser accepts a bare JSON object, a fenced JSON object, or the first balanced object followed by prose. An unparseable response is logged as an episode error; there is no semantic re-prompt in this LIBERO path. A parsed response is converted to intervals as follows:

1. Clamp the start and end ratios to $[0, 1]$. If an end does not exceed its start, replace it by $\min(1, \text{start ratio} + 1/K)$.
2. Set the first start to 0. For every later phase, discard its proposed start as an independent boundary and set it to the previous final end. Before snapping, a non-final end is $\text{round}(\text{end_ratio} T)$; the final end is T .
3. Interpret each boundary using the *next* description. Before Release/Open it may snap only to a gripper-open event; before Grasp/Close only to gripper-close; otherwise only to a velocity peak or stillness event. The nearest compatible hint is used only within radius $\max(5, \lfloor T/30 \rfloor)$.

Local source	Fields read	Use before or after the VLM call	VLM-visible?
episodes.jsonl	episode_index, length, task or tasks[0]	Selects the episode and its single authoritative instruction. A supplied episode list may override the selected task string.	Task only
info.json	Video features and action names	Selects the first available exterior/non-wrist video when possible and recognizes the seven-dimensional LIBERO action layout.	No
Episode Parquet	frame_index, action	Rows are sorted by frame. The first six action dimensions yield an action-space motion magnitude; the seventh yields gripper events. Raw values, units, and thresholds stay local.	Derived hints only
Episode video	Selected RGB frames	Twelve frames are resized to 144 pixels in height, concatenated horizontally, and JPEG-encoded at quality 92. The full video is not uploaded.	Contact sheet only

Table S1: Inputs used for automatic phase annotation. “Derived hints” contain an event type and time but not the underlying action vector. Because the action dimensions have no stated SI units, motion is reported in action space.

4. Reserve at least one frame for every remaining phase and prefer a three-frame minimum when feasible. Renumber phases and enforce contiguous half-open coverage through frame T .

Descriptions are stripped and truncated at 120 characters. Missing confidence defaults to 0.85. The field `boundary_evidence=vlm_vision+action_snap` records the postprocessing path; by itself, it does not mean that a boundary was snapped to a nearby hint.

A.5 Deterministic Place–Release Reconstruction

Release receives a second, action-only refinement because sparse images do not reliably identify the opening instant. Any adjacent VLM Place–Release pair is first collapsed back to its combined placement window. Place candidates are recognized from `place`, `put`, `insert`, `set`, `drop`, `hang`, or `stack`. The postprocessor then re-detects the complete, uncapped sequence of stable gripper-open events from the raw local action stream.

For a Place interval, a candidate event must be unused, occur no earlier than its start minus r , and occur no later than its end plus r or the next Place start, whichever comes first, where $r = \max(5, \lfloor T/30 \rfloor)$. The last eligible event in this window is chosen. Place ends at that frame; Release starts there and ends no later than the original placement end or the next stable close. It must include the three-frame open confirmation, borrowing frames from the following interval when necessary. The reconstructed segment is assigned a stored score of 1.0, a valid audit flag, and evidence of the form `stable_gripper_close_to_open@F...:confirm=3`.

Each ordered Place must use a distinct later release event. A task that requires placement but has no Place description, lacks usable action data, or cannot form a consistent interval is excluded and logged. Two recoverable cases—a missing Release for a detected Place and a coarse multi-object Place description—retain the episode but set the stored scores of that Place and all later segments to zero, with `phase_valid=false`. If motion continues after a final verified Release and the gripper closes again, the remaining

tail is likewise stored as a completion segment with score zero and a false audit flag. Only Release stages matched to an action event are retained.

A.6 Stored Annotation Record

The stored episode includes the selected task, final segments, annotation scores and audit flags, hint frames and types, generator metadata, and the terminal-release audit. The stored episode score is the mean of its segment scores. Fatal cases enter a separate error log, while recoverable fallback cases remain in the annotation record.

Stage A uses a binary sample mask β_i . The loader requires contiguous stage coverage and checks the recorded episode length against action-padding metadata when available. A sample receives $\beta_i = 1$ if these checks pass and its action chunk has at least one non-padded step, and zero otherwise. Stored scores and `phase_valid` flags do not enter this mask. Consequently, a zero-score or audit-flagged stage can still contribute to the occupancy mixture and the training loss.

Suite	Episodes	Queries	Pure	Mixed	Invalid
Spatial	434	2,367	1,173	1,194	7
Object	457	2,457	1,377	1,080	3
Goal	433	1,928	1,088	840	1
Long	388	2,755	1,674	1,081	115
Total	1,712	9,507	5,312	4,195	126

Table S2: Annotation and query composition across the four ten-task LIBERO suites. One query is created at each phase start. “Invalid” counts stored `phase_valid=false` labels.

B Chunk-Aligned Query Construction

B.1 Occupancy Labels

For a query at frame t , let $\mathcal{C}_t$ contain the valid, non-padded action indices in the next $H = 32$ demonstration steps and

Statistic	Value
Annotated phases / phase-start queries	9,507
Exact unique phase descriptions	503
Phases per episode, mean / median	5.553 / 5
Phases per episode, min / max	2 / 11
Valid / invalid flags	9,381 / 126
Duration in frames, mean / median	29.21 / 23
Duration in frames, 10th / 90th percentile	3 / 59
Queries with action padding	2,714

Table S3: Aggregate annotation statistics from the 9,507-row phase-start manifest.

let g_τ denote the phase containing step τ . The occupancy of phase k is

$$w_{t,k} = \frac{|\{\tau \in \mathcal{C}_t : g_\tau = k\}|}{|\mathcal{C}_t|}.$$

All intersecting phases are retained, and occupancies are renormalized over the valid steps. Consequently, a short final chunk still sums to one and does not assign semantic mass to padded actions. Of the 9,507 phase-start queries, 2,714 contain action padding. A valid final chunk retains unit sample weight; its shorter length changes the occupancy denominator, not its loss weight.

Each phase contributes one query at its start frame. A query is *Pure* when one stage occupies all valid future steps and *Mixed* when two or more stages overlap the chunk. Mixed-2 contains exactly two stages; Mixed-3+ contains at least three. These categories use the full valid action horizon, not only the stage at the first step.

B.2 Fixed Four-Token Teacher

Each free-form description is encoded by the frozen UMT5 encoder (Chung et al. 2023). Its valid token sequence is partitioned into four contiguous, non-empty intervals and mean-pooled within each interval. We center the pooled training-description vectors and construct a fixed random projection from a seed-0 Gaussian matrix followed by reduced QR decomposition; its 128 orthonormal rows are independent of the semantic training data. The teacher target is the occupancy-weighted mixture of the projected stage slots:

$$T_{t,n} = R \left(\sum_k w_{t,k} \bar{h}_{k,n} - \mu \right), \quad n \in \{1, \dots, 4\}.$$

Thus $T_t \in \mathbb{R}^{4 \times 128}$ preserves coarse token order while matching the temporal support of the action chunk. A valid sample must supply all four pooled teacher slots; missing or incomplete teacher representations raise an error.

C Teacher Matching

C.1 Episode-Local Retrieval

The analysis uses all 1,712 annotated LIBERO training demonstrations and the 10,000-step CASD generator checkpoint. It measures representation matching on training episodes, separately from the rollout success rates in the main paper.

For an episode e containing N_e annotated phases, the gallery contains the N_e fixed teacher targets constructed at those phase starts. The CASD generator predicts P_i from the current observation, robot state, and task instruction. Phase boundaries and future occupancies are used only after prediction to construct the targets and gallery candidates.

We use the negative normalized Stage-A regression distance as the matching score:

$$s(i, j) = -\frac{\|P_i - T_j\|_F^2}{4 \cdot 128 \cdot \sigma_T^2},$$

where σ_T is the RMS teacher scale stored with the fixed teacher. The correct item is the teacher context at the same phase start. If multiple candidates tie for the maximum score, Recall@1 assigns $1/m$ credit when the correct candidate is among the m tied maxima and zero otherwise.

For random guessing, query i has expected recall $1/N_e$. Chance Recall in Table S4 is the average over queries; each episode contributes in proportion to its number of queries.

Subset	Queries	Share	Chance	Recall@1
Pure	5,312	55.87%	0.185	0.430
Mixed	4,195	44.13%	0.174	0.765
Mixed-2	3,643	38.32%	0.175	0.761
Mixed-3+	552	5.81%	0.166	0.790

Table S4: Episode-local teacher matching at annotated training phase starts. Mixed-2 and Mixed-3+ partition Mixed. Values are averaged over queries, with fractional credit for top-score ties.

The corresponding Recall@1-to-chance ratios are 2.3, 4.4, 4.4, and 4.8 for Pure, Mixed, Mixed-2, and Mixed-3+, respectively, computed before rounding. Mixed queries cross stage boundaries within the 32-step action horizon. The ratios also depend on gallery composition and do not measure the relative difficulty of the subsets.

D Training Configuration

For LIBERO, the CASD generator has hidden width 128. Its inputs comprise a 432-dimensional frozen-VAE feature of the current frame, the eight-dimensional robot state, and frozen UMT5 task embeddings. The teacher has four tokens of width 128 and uses occupancy over the valid part of an $H = 32$ action chunk.

Stage A is configured for 10,000 optimization steps, a global batch size of 128, learning rate 10^{-4} , weight decay 10^{-2} , and a cosine schedule. The direct-regression and same-task hardest-negative ranking losses each have coefficient 1. The minimum eligible teacher distance is 0.05, and the margin ratio is 0.5. Only CASD generator parameters are optimized. The frozen VAE, UMT5 encoder, teacher transformation, and policy backbone receive no Stage-A updates. Stage A samples frame anchors throughout the annotated training demonstrations and applies the binary mask defined above. The 9,507-row phase-start manifest is used only for retrieval analysis; stored annotation flags do not filter this manifest either.

At Stage B, the selected CASD generator checkpoint is loaded and frozen. Policy training recomputes its prediction from the current observation rather than reading teacher tokens or cached future-dependent occupancies. This keeps the CASD-token source identical during policy training and deployment. The direct-regression and ranking losses are disabled in Stage B.

Stage-B injection. Fast-WAM-Joint (Yuan et al. 2026) contains 30 aligned video/action expert blocks of widths 3072/1024. Each expert independently projects the four shared predicted tokens to width 1024 with an affine map and GELU. The resulting context is supplied to an independent rank-192 Q/K/V/O cross-attention adapter in every video and action block. These adapters use 24 heads of dimension 128. For expert width d_e , Q maps $d_e \rightarrow 192 \rightarrow 3072$, K/V map $1024 \rightarrow 192 \rightarrow 3072$, and O maps $3072 \rightarrow 192 \rightarrow d_e$. Thus the attention width is $24 \times 128 = 3072$ in both experts, even though the action residual stream has width 1024. The last linear layer of O is zero-initialized so the pretrained function is unchanged at initialization.

E Policy Metrics and Result Sources

E.1 Success Aggregation

For a task with n evaluation episodes and binary outcomes y_j , success is $100 \sum_j y_j / n$. LIBERO reports the mean task success within each ten-task suite and the mean of the four suite scores. RoboTwin reports the mean over 50 tasks separately for Clean and Rand.; its Average gives the two conditions equal weight. The MolmoSpaces comparison reports the four manipulation categories Pick, Pick-and-Place, Open, and Close, with equal weight in the displayed average.

LIBERO-Plus uses a different aggregation. Let n_c and s_c denote evaluated episodes and successes in perturbation category c . Its pooled score is

$$S_{\text{pooled}} = 100 \frac{\sum_{c=1}^7 s_c}{\sum_{c=1}^7 n_c}.$$

Categories have different sizes, so this is not an unweighted mean of the seven displayed percentages. Table S5 gives the totals behind the CASD rows. The Joint run’s suite-summary value of 72.36 uses a different aggregation and is not the pooled value in the main table.

Fast-WAM model	Episodes	Successes	Pooled (%)
Uncond+CASD	10,030	5,737	57.2
Joint+CASD	10,030	7,238	72.2
IDM+CASD	10,030	7,417	73.9

Table S5: LIBERO-Plus aggregation from the seven category counts.

Policy scores in the main paper use one decimal place. Differences between our LIBERO variants and their references use the unrounded run averages: IDM+CASD is 98.85 and Uncond+CASD is 94.45, giving changes of +0.85 and −3.15 percentage points. The main text rounds these changes to +0.9 and −3.2 points.

E.2 RoboTwin Per-Task Results

Table S6 reports success rates for all 50 RoboTwin tasks using the Uncond, IDM, and Joint Fast-WAM variants with CASD under Clean and domain-randomized (Rand.) evaluation. Our policies execute the first 28 actions of each predicted chunk before replanning from a new observation (`replan_steps=28`). Each task has equal weight within a condition, and Clean and Rand. have equal weight in the overall average. The overall scores are 90.92%, 92.39%, and 92.96% for Uncond+CASD, IDM+CASD, and Joint+CASD, respectively, matching the rounded entries in the main paper. Unaugmented reference results come from the published sources described below.

E.3 Published References

The unaugmented Fast-WAM rows in Tables 2 and 4 follow the official report (Yuan et al. 2026) for the same named variants and benchmark settings. Table 3 uses external Fast-WAM checkpoints evaluated by RIFT (Appendix D, Table 5, version 2) (Zhang et al. 2026); other baselines follow the LIBERO-Plus report (Fei et al. 2026).

Sources for the remaining LIBERO and RoboTwin models appear in the main paper. MolmoSpaces references use the official September 2, 2026 leaderboard snapshot (Allen Institute for AI 2026) and were not re-evaluated here.

F Paired Rollout Analysis

The Figure 2 rollout outcomes, action traces, videos, and CASD generator predictions come from zero-indexed trial 2 of zero-indexed LIBERO-Long task 3: put the black bowl in the bottom drawer of the cabinet and close it. Fast-WAM-Joint succeeds in 49/50 trials and Fast-WAM-Joint+CASD in 50/50; trial 2 is the paired failure/success case. Their recorded initial observations and all 22 common inference seeds match. The displayed transition is r15→r16, from Move/place toward Close.

Behavioral signals. CASD-token change is the symmetric nearest-neighbor cosine distance between adjacent 4×128 token sets. Motion divergence is the root-mean-square difference over the first six non-gripper action coordinates, aligned within the same replan and restricted to executed actions. Gripper mismatch is the fraction of aligned gripper-sign commands that disagree. Token change and motion divergence are each divided by their maximum over the 22 common replans; gripper mismatch remains on its native $[0, 1]$ scale.

Semantic fit. For each task-specific stage description u_k , let U_k be its fixed, four-slot projected teacher prototype. For visualization, we fit a student context P_t by solving

$$\min_{\alpha \geq 0, \mathbf{1}^\top \alpha = 1} \left\| P_t - \sum_k \alpha_k U_k \right\|_F^2.$$

The implementation uses deterministic projected gradient descent on the simplex. Prototype weights are then summed into the displayed semantic categories. These are reconstruction coefficients, not annotated occupancies or calibrated probabilities. They are computed after the rollout and are never supplied to the policy.

Frame selection. The eight frames show both policies before the boundary, inside the r16 divergence chunk, in the following replan, and at the CASD completion step. The manifest records the exact frame indices and selection rationale. The figure connects a semantic transition with behavior in one selected rollout, rather than measuring a causal effect of changing an individual token.

G Implementation Scope and Reproducibility

The implementation described here covers the Fast-WAM annotation, teacher-construction, training, and evaluation pipeline. The separate DreamZero integration is outside this implementation description.

The source archive accompanying this preprint contains the manuscript and final figures. It does not include training or evaluation code, model weights, generated phase annotations, normalization statistics, embedding caches, per-query retrieval results, or full rollout traces. The archive is therefore not a self-contained reproduction of the reported experiments.

For a complete teacher-matching record, the analysis scripts export a row per query with suite, task, episode, phase start, validity, padding, overlapping stages, and occupancies, plus per-query tie credit. Counts and chance baselines can be reconstructed from such a manifest; Recall@1 additionally requires the corresponding predictions or checkpoint and teacher artifact. Checkpoint reuse also requires the matching model configuration and normalization artifacts.

Annotation variability. The LIBERO launcher requests the `gpt-5.4` alias through an OpenAI-compatible service. The saved annotations, rather than new service responses, are the inputs to subsequent training. Action-based snapping refines boundary timing but does not verify every free-form description. Stored annotation scores are heuristic, and no inter-annotator agreement is reported.

Task	Uncond+CASD		IDM+CASD		Joint+CASD	
	Clean	Rand.	Clean	Rand.	Clean	Rand.
adjust_bottle	100	100	93	100	99	99
beat_block_hammer	97	98	100	96	99	95
blocks_ranking_rgb	100	98	100	99	100	99
blocks_ranking_size	88	88	83	84	78	92
click_alarmclock	100	100	100	100	100	100
click_bell	100	100	100	100	100	100
dump_bin_bigbin	96	100	98	98	95	95
grab_roller	100	100	100	100	100	100
handover_block	92	85	98	92	99	91
handover_mic	99	99	98	99	100	100
hanging_mug	38	49	49	54	42	49
lift_pot	100	100	100	100	100	100
move_can_pot	84	89	100	98	97	94
move_pillbottle_pad	98	99	99	99	100	100
move_playingcard_away	100	100	100	99	100	100
move_stapler_pad	72	69	77	82	83	78
open_laptop	98	100	98	98	95	99
open_microwave	58	70	70	72	85	93
pick_diverse_bottles	83	85	89	87	90	92
pick_dual_bottles	93	91	100	98	100	99
place_a2b_left	92	95	93	97	93	98
place_a2b_right	96	95	96	96	97	93
place_bread_basket	92	94	90	98	91	97
place_bread_skillet	89	96	93	87	87	92
place_burger_fries	95	99	99	99	100	98
place_can_basket	72	72	75	67	71	72
place_cans_plasticbox	97	96	98	98	98	99
place_container_plate	96	99	98	98	98	100
place_dual_shoes	96	91	88	85	88	92
place_empty_cup	98	100	100	100	100	100
place_fan	99	98	99	95	98	95
place_mouse_pad	82	82	95	94	93	92
place_object_basket	90	87	86	78	85	78
place_object_scale	96	94	90	95	97	94
place_object_stand	94	91	98	97	95	95
place_phone_stand	95	96	99	96	99	96
place_shoe	95	99	94	97	96	95
press_stapler	95	98	89	89	89	91
put_bottles_dustbin	90	95	93	97	92	96
put_object_cabinet	92	88	95	87	84	85
rotate_qrcode	87	87	88	75	87	88
scan_object	88	88	95	96	91	96
shake_bottle	100	100	100	100	100	100
shake_bottle_horizontally	100	100	100	100	100	99
stack_blocks_three	91	94	98	96	98	94
stack_blocks_two	100	99	100	100	100	100
stack_bowls_three	82	78	75	85	80	83
stack_bowls_two	94	95	89	95	92	95
stamp_seal	83	81	86	89	95	91
turn_switch	52	61	77	70	77	84
Mean (50 tasks)	90.48	91.36	92.56	92.22	92.66	93.26
Overall (Clean/Rand.)	90.92		92.39		92.96	

Table S6: RoboTwin 2.0 per-task success rates (%) for the three CASD-augmented Fast-WAM variants. Clean and Rand. correspond to the two evaluation conditions in the main paper. Mean values are recomputed from all 50 task rows; Overall averages the two condition means.